



VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2015
Higher 2

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CANDIDATE NAME

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CLASS	1	4	S		
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EXAM NUMBER							
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BIOLOGY

9648/03

Applications Paper and Planning Question

Paper 3

18 September 2015

Additional Materials: Answer Booklet/ Paper

2 Hours

READ THESE INSTRUCTIONS FIRST

Write your Class, exam number and name on all the work you hand in.

Write in dark blue or blue pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use any staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

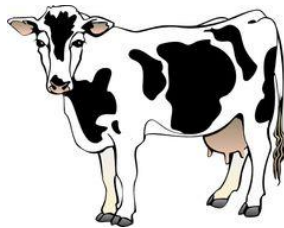
At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
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5	
Total	

Answer **all** questions.

- 1 One of the first potential targets for biotechnology products for animal products was the use of bovine somatotropin (bST) in cows. bST influences the tissue metabolism so that more nutrients can be channelled into milk synthesis. This allows the animals to achieve an increased milk yield while remaining normal and healthy. bST is widely marketed to dairy farmers in the US, who can add the hormone as a form of supplement to the diet of their cows.



A student wanted to carry out an experiment to clone the *bST* gene for expression. **Fig.1.1** below shows a map of the region of the cow's genome where the *bST* gene is located and **Fig. 1.2** shows the plasmid that he intends to use as a vector.

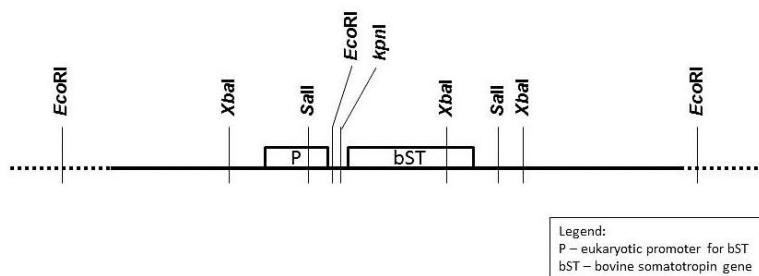


Fig.1.1

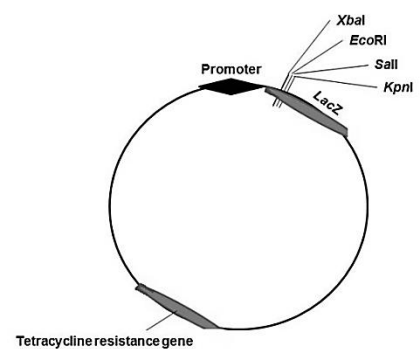


Fig. 1.2

- (a) (i) With reference to **Fig. 1.1** and **Fig. 1.2**, state and explain the most ideal combination of restriction enzymes that he would use to isolate the *bST* gene for expression.

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(ii) Describe how two features of the plasmid shown in **Fig. 1.2** would allow him to identify the bacteria that have taken up the **bST** gene.

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(iii) He cultured the bacteria containing the **bST** gene in an incubator but no hormone was produced. Explain how this problem can be solved.

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(b) Commercial production of bST makes use of fermenters containing cultures of cells expressing the gene.

Suggest two reasons why this method of producing bST is preferred over insertion of **bST** gene directly into the genome of the cow.

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- (c) With the advancement in genetic engineering techniques, more food has been genetically modified. Although the US Food and Drug Administration have stated that such food is safe for human consumption, some consumers are demanding that such food be labelled as genetically modified (GM) food.

Discuss briefly one ethical concern about not labelling genetically modified food.

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[Total: 12]

- 2 Tissue culture is a technique that can be used to propagate plants that are difficult to grow from seeds. Explants are removed and cultured in nutrient rich medium to produce calluses.

(a) (i) Define the term *callus* and explain how its features have been exploited in tissue culture.

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(ii) Suggest two advantages of tissue culture over the more traditional methods of cloning plants, such as taking cuttings or grafting.

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(b) Nutrient media formulations for tissue culture must include:

- a source of carbon
- various essential minerals
- vitamins
- growth regulators

The rates of growth of rattan explants in two types of regulators are shown (**Fig. 2.1**).

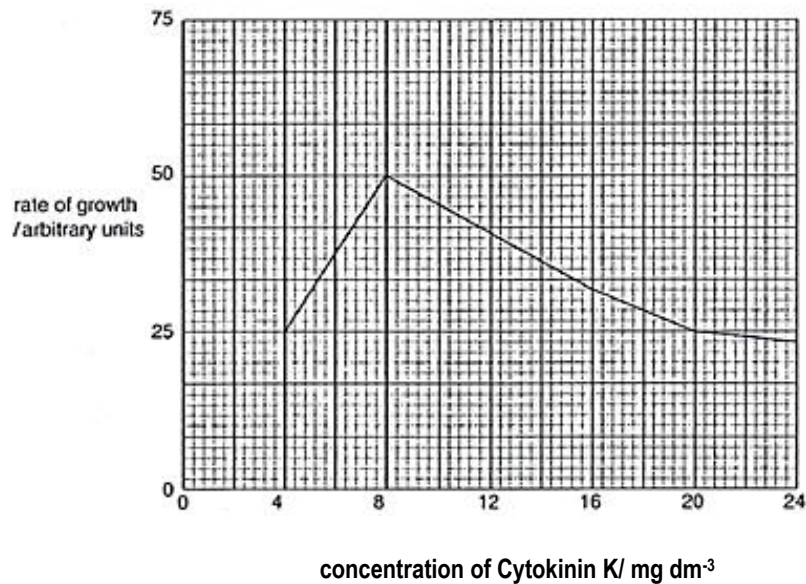
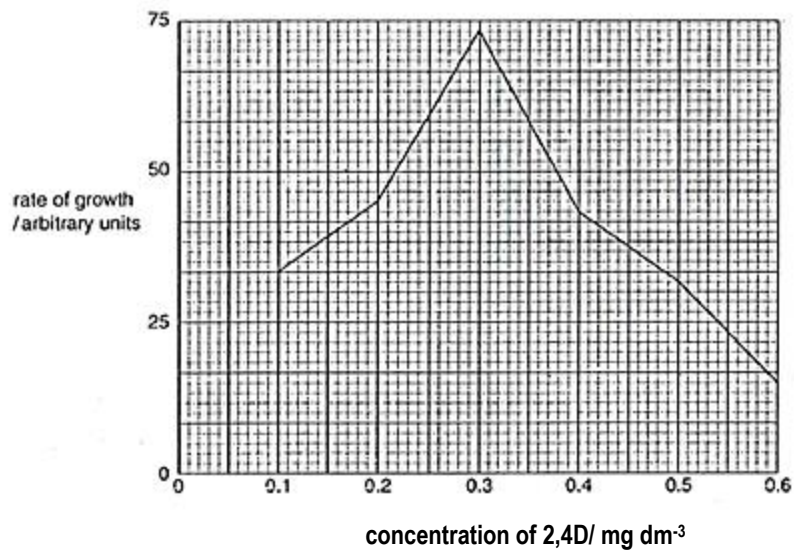


Fig. 2.1
Adapted from N07 A level

- (i) With reference to **Fig. 2.1**, compare the extent to which variations in the concentrations of **2,4 D** (a synthetic auxin) and **cytokinin K** affect the successful growth of the explant.

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- (ii) **2,4 D** has been used to control the growth of weeds because it is also able to cause the death of cells. Use the information in **Fig. 2.1**, to justify what might be the lowest effective concentration of **2,4 D** to elicit cell death.

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(c) Nitrogen is also a vital constituent of tissue culture medium. During the investigation, it was determined that ammonium nitrate is a better source of nitrogen than potassium nitrate for the growth of the explants.

- (i) Explain why the explants need a source of nitrogen.

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(ii) Suggest why ammonium nitrate may be a better source of nitrogen than potassium nitrate.

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[Total: 14]

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- 3 Many marine fish species that inhabit icy sea waters synthesize antifreeze proteins (AFP) to protect them from freezing. The *AFP* gene codes for the antifreeze protein in the ocean pout. A transgenic Atlantic salmon line is created by placing the Pacific chinook salmon growth hormone (GH) cDNA driven by the AFP promoter that was taken from the ocean pout. This transgenic salmon is called AquAdvantage salmon.

The ocean pout antifreeze promoter is constitutively (i.e. continually) expressing in all tissues of the transgenic salmon. In contrast, the growth hormone promoter in standard salmon is only active in response to certain environmental cues such as day length and temperature in some tissues.

The growth of AquAdvantage salmon and the wild type (standard) salmon was measured after the first feeding and the results are shown in **Fig. 3.1**.

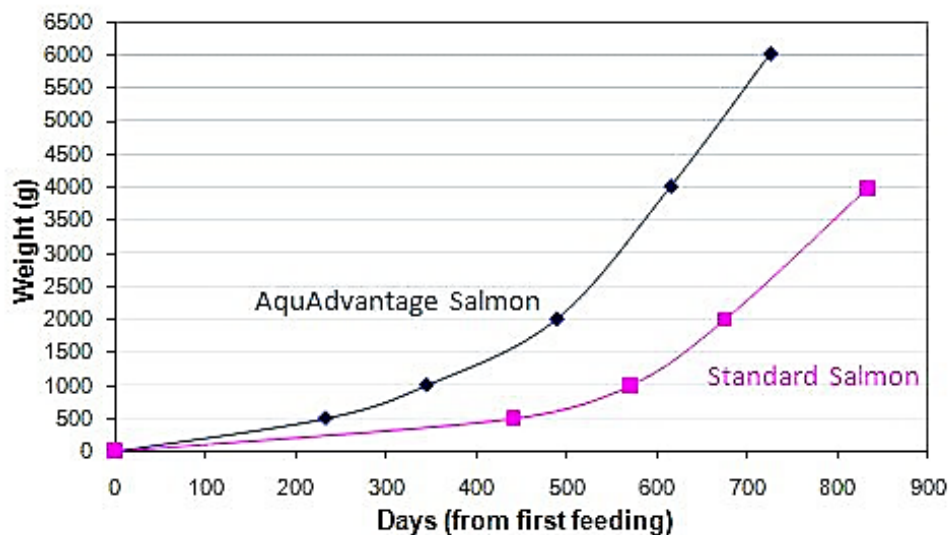


Fig. 3.1

- (a) Use the context of AquAdvantage salmon to explain the term ***genetically modified organisms***.

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- (b) (i) Based on **Fig. 3.1**, and the information provided, describe the growth of transgenic salmon compared to standard salmon.

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[2]

- (ii) Relate the possible effect of the AFP promoter on growth rate observed in transgenic salmon as compared to standard salmon.

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[2]

The development of the AquAdvantage Salmon began in 1989 with the micro-injection of the transgene into fertilized Atlantic salmon eggs.

Fig. 3.2 shows the structure of the transgene. It contains the coding sequence of the growth hormone gene from the Pacific Chinook salmon under the control of 5' UTR, promoter and 3' UTR sequences of the antifreeze *OP5a* gene from the ocean pout.

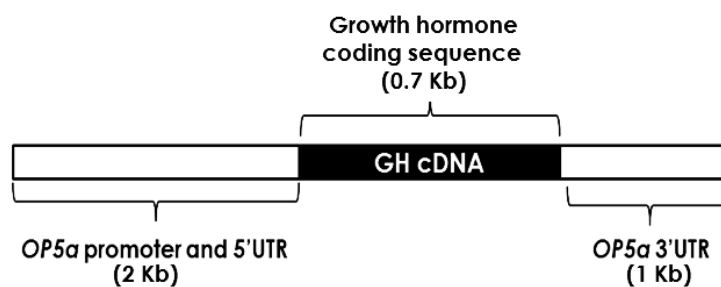


Fig. 3.2

- (c) Using the information provided, outline briefly how one would use PCR to distinguish between a transgenic salmon and a standard salmon.

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The AquAdvantage salmon is produced as a triploid and therefore sterile. Transgenic eggs are generated in the AquaBounty Technologies facility on Prince Edward Island (Canada) before shipping to a restricted facility in Central America, where they will be reared. Once they reach market size (**Fig. 3.1**), they will be harvested for processing (e.g. preparation of fish fillets, steaks, etc.) for consumption. No live fish will be sold to consumers.

- (d) Identify one social and one ethical concern surrounding genetically modified organisms and discuss whether AquAdvantage salmon successfully addresses these concerns.

Social concern

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Ethical concern

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[Total:14]

Name: _____

CT Group: _____

Mark: _____

4 Planning question

You are required to plan but not carry out, an investigation into the effect of different wavelengths of light on the rate of photosynthesis in two varieties of the same species of a garden plant. These two varieties differ in their ability to produce chlorophyll b.

Variety **A** Normal wild type (presence of chlorophyll b)

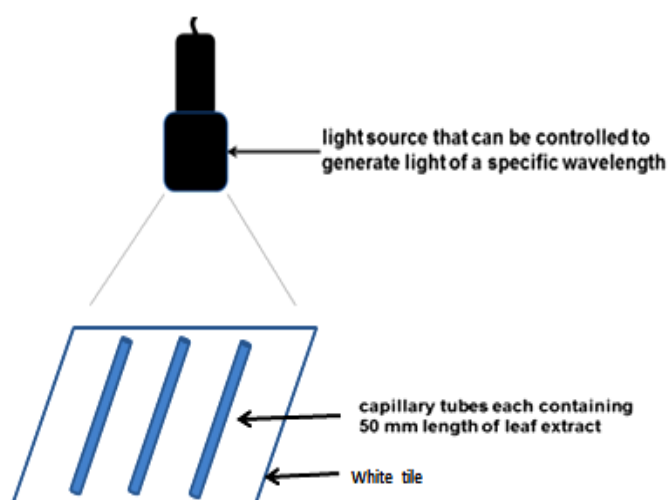
Variety **B** Chlorophyll b deficient

Leaf extracts from each variety are prepared by grinding 10 g of the leaf sample in 30 cm³ of ice cold phosphate buffer solution. The suspension is then filtered through a fine mesh and the filtrate covered with aluminum foil and placed in an ice bath.

You are provided with thin capillary tubes that can be used to draw up the filtered leaf extract.

The method used to find out the rate of photosynthesis is to determine the time taken for blue DCPIP added to the filtrate to decolourize. When oxidized, DCPIP is blue in colour and it turns colourless when it accepts electrons and becomes reduced.

Figure 4.1 below shows the setup for the experiment.



The Visible Light Spectrum	
Color	Wavelength (nm)
Red	625 - 740
Orange	590 - 625
Yellow	565 - 590
Green	520 - 565
Cyan	500 - 520
Blue	435 - 500
Violet	380 - 435

Figure 4.1

Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use.

- Filtered leaf extracts from the two varieties in beakers covered with aluminum foil (sitting in ice)
- A bench light source that is fitted with refractors and filters to enable you to get a specific wavelength known as monochromatic light
- 1% DCPIP solution
- Capillary tubes (50 mm in length)
- Timer e.g. stop watch or stop clock
- Petri dishes
- White tile
- Ruler
- Aluminum foil
- Glass rods
- Teat pipettes

You can also select from the following apparatus:

- A variety of different sized beakers, measuring cylinders or syringes for measuring volume

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram(s) if necessary,
- identify the independent and dependent variables,
- include full details and explanations of the procedures that you would adopt to ensure that the results obtain will be as quantitative, precise and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms.

[Total:12]

[illegible]

[illegible]

5 Free-response question

Write your answers to this question on separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
 - must be in continuous prose, where appropriate,
 - must be set out in sections (a), (b) etc., as indicated in the question.
- (a) Describe how a non-viral gene delivery method can be used to treat Cystic Fibrosis. [6]
- (b) Outline the process of nucleic acid hybridisation and explain how it can be used to detect and analyse restriction fragment length polymorphism (RFLP). [6]
- (c) Discuss the goals and benefits of the Human Genome Project. [8]

[Total: 20]